Ascent Performance of the Rockot/Breeze-KM Launch Vehicle from Plesetsk

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Abstract. A three-degree-of-freedom ascent model of Rockot/Breeze-KM is calibrated against the reference mission of the User's Guide [1]: 1,570 kg to a 650 km circular orbit at 86.58°. Stage-1 and stage-2 burnout speeds agree within 0.2 % and 0.5 %. Maximum dynamic pressure is 64.8 kPa and peak axial load 7.2 g. The upper stage inserts into a 643 × 666 km orbit at 86.34° with 235 kg of propellant remaining.

1 Vehicle

Engine, specific impulse and burn-time data are from [1], dry masses from [2]. Stage-1 and stage-2 propellant loads are fitted (Section 4); thrust follows from load, burn time and specific impulse. Lift-off mass is 117.0 t and lift-off thrust-to-weight ratio 1.75.

Table 1. Stage parameters. Ideal Δv totals 10,061 m/s (fairing carried to stage-2 burnout).

	Stage 1	Stage 2	Breeze-KM
Engines	RD-0233 ×3, RD-0234	RD-0235, RD-0236 ×4	S5.98M
Thrust, vacuum (kN)	2,178	248	20
I_{sp}, vacuum (s)	310.0	320.0	325.5
I_{sp}, sea level (s)	285.0	–	–
Burn time (s)	121	183	≤ 794
Propellant (kg)	86,691	14,475	4,975
Dry mass (kg)	5,695	1,485	1,320
Ideal Δv, vacuum (m/s)	4,109	2,790	3,161

2 Mission performance

Table 2. Reference mission, Plesetsk (62.8° N, 40.3° E).

	Model	User's Guide [1]
Payload (kg)	1,570	1,570
Lift-off mass (t)	117.0	107
Max. dynamic pressure (kPa)	64.8 at 51 s	61.3 at 50 s
Peak axial load, stage 1 (g)	7.2	6.2–8.0
Peak axial load, stage 2 (g)	2.7	≤ 3.0
Peak axial load, Breeze-KM (g)	0.64	–
Transfer orbit (km)	160 × 635	160 × 635
Final orbit (km)	643 × 666	650 × 650
Inclination (deg)	86.34	86.58
Lift-off to transfer orbit (s)	1,038	882
Breeze-KM propellant remaining (kg)	235	–

3 Trajectory

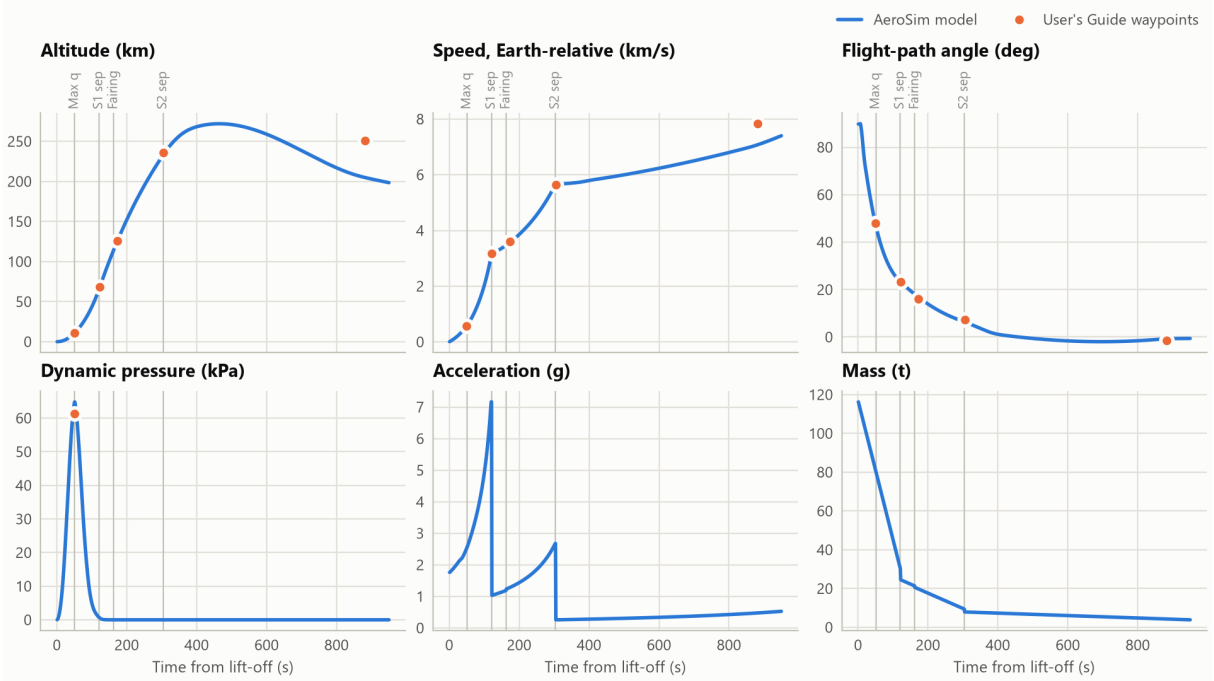


Figure 1. Ascent to the end of Breeze-KM burn 1. Line: model; points: waypoints of [1, fig. 3-6]. Vertical lines mark max. *q*, stage-1 separation, fairing jettison and stage-2 separation.

Table 3. State at flight events: model against [1, fig. 3-6]. *v*: Earth-relative speed; γ: flight-path angle.

Event	*t* (s)		*h* (km)		*v* (m/s)		γ (deg)	
	Model	Guide	Model	Guide	Model	Guide	Model	Guide
Max. *q*	51	50	10.9	10.7	591	567	46.9	47.9
Stage-1 sep.	121	122	65.9	68.3	3,156	3,163	23.3	23.3
Fairing jettison	162	173	114.2	125.6	3,464	3,599	18.0	16.0
Stage-2 sep.	304	305	233.0	235.7	5,663	5,634	6.5	7.3
Burn-1 cutoff	1,038	882	189.9	250.7	7,902	7,832	−0.6	−1.5

4 Calibration and limitations

Stage-1 and stage-2 propellant loads, pitch-kick angle, stage-2 pitch rate and the Breeze-KM climb-rate law were fitted by least squares to ten waypoints of [1]; eight are met within tolerance. The two misses, lift-off mass (+9.3 %) and burn-1 cutoff (+156 s), indicate that the model needs more energy than the flight vehicle, consistent with its simplified insertion steering and generic drag curve. The model is a point mass with constant thrust per stage, J_2 gravity and an exponential upper atmosphere [3].

References

[1] Eurockot Launch Services, *Rockot User's Guide*, EHB0003, Issue 5, Rev. 0, 2011.

[2] M. Wade, "Rokot," *Encyclopedia Astronautica*, astronautix.com.

[3] D. A. Vallado, *Fundamentals of Astrodynamics and Applications*, table 8-4.